

Devis Saputra

Learning Designer | AIEd

Medan, Indonesia | Open to remote work and relocation

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RESEARCH PROFILE

Learning Designer with 10+ years across higher education, corporate learning and applied technology, extending this experience into Artificial Intelligence in Education (AIEd). Established expertise in instructional design, curriculum development, Moodle learning systems and content governance; growing AI engineering practice through Python, machine learning and reproducible evaluation projects. Combines a BEng in Informatics, an MBA in International Business and seven years of university teaching with applied research and learning-system implementation. Research direction: how AI-supported feedback, learner modeling and decision support can improve learning while preserving learner agency, teacher judgment and credible evaluation.

EDUCATION

MBA, International Business | October 2012 to June 2014

THWS Würzburg-Schweinfurt, Germany | Final grade: 1.8 (German scale)

Thesis: Leadership on Flat Organizational Structure: Focusing on Organization Using Self-Managing Team and Open Allocation Strategy. Qualitative comparative analysis of Valve and GitHub; thesis record dated 2013, degree completed 2014.

BEng, Informatics | August 2004 to March 2010

Institut Teknologi Bandung, Indonesia

Thesis: Image Retrieval System Using Geotagged Information. Combined JPEG EXIF geographic metadata with text retrieval; evaluated TF, IDF and TF-IDF weighting using precision, recall and average precision.

SELECTED INDEPENDENT RESEARCH AND AI PROJECTS

Public code and working papers, 2026. These projects demonstrate developing research and engineering practice; they are distinct from the published outputs listed on page 3.

Knowledge Tracing Benchmark

Evaluates prior, Bayesian, logistic and GRU baselines on ASSISTments 2009, with 4,148 learners and 274,331 interactions. Learner-disjoint partitions and prediction-before-update logic support inspection of leakage and generalization.

Misconception-Aware Retrieval Evaluation

Compares retrieval approaches against external misconception labels and reports paired retrieval differences. Connects information retrieval to the quality of evidence used in educational feedback.

Probability Calibration

Studies probability quality on UCI Bank Marketing using uncalibrated and calibrated logistic regression across five fixed holdouts. Separates discrimination from calibration and explicitly examines call-duration leakage in prospective use.

Decision Value of Training Evidence

Reanalyzes the 445-participant randomized National Supported Work subset. Reports effect uncertainty and an illustrative decision-threshold frontier, separating evidence of benefit from a business decision rule.

Technology Skill Renewal Pressure

Compares O*NET releases across 902 common occupations to study software-skill persistence and sensitivity to category definitions. Frames workforce reskilling questions without treating database turnover as measured worker skill loss.

Portfolio spans learning and development, AI in education, AI engineering and systems research. Repository documentation records data provenance, calculation definitions and interpretation limits.

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Freelance Instructional Design & Learning Technology

Independent consulting | January 2025 to present

Remote, Indonesia

- Design role-specific learning structures and synthesize research into instructional content and learning-technology deliverables.
- Evaluate AI model outputs for response quality, domain accuracy and safety alignment within human-feedback workflows.

Learning and Development Manager

Suitmedia | February 2023 to August 2024

Jakarta, Indonesia; remote

- Led Rocketiversity, a customized Moodle learning system, through content audit, course structuring, manager approval and rollout with a four-person project team.
- Established 120 SCORM-compliant courses and structured onboarding for around 100 employees; the project case study reports a 30% reduction in onboarding time.
- Converted fragmented departmental materials into reusable learning units, clarified content ownership and assigned trainers from the teams responsible for the subject matter.

Head of Engineering Management Program

Institut Teknologi Del | November 2017 to May 2021

Laguboti, Indonesia; concurrent with lecturer appointment

- Led the 2019 to 2024 curriculum revision, translating expert consultation, graduate feedback and industry needs into a 144-SKS, eight-semester curriculum.
- Aligned learning outcomes, course sequencing and assessment; coordinated peer review, Academic Senate approval and course-equivalency rules for student progression.

Lecturer in Engineering Management

Institut Teknologi Del | May 2016 to June 2023

Laguboti, Indonesia

- Taught engineering economics, entrepreneurship, technology management and related management subjects, using applied problems and project-based assessment.
- Supervised more than 50 undergraduate theses and capstone projects; developed internship guidance, assessment structures and practical-work documentation.
- Contributed to the Astra Lean Production System teaching laboratory and co-designed practicum activities that connected lean-production concepts with hands-on learning.

Program Manager, Health Technology Initiative

Summit Institute for Development | May 2015 to October 2015

Mataram, Indonesia

- Directed four developers and six field workers across 16 districts for the THRIVE/OpenSRP mobile health workflow initiative.
- Used health-worker interviews and a field timing study to standardize form completion and diagnose XLSForm latency; an interim form-splitting solution reduced most form loads from a baseline range of 20 seconds to 20 minutes to under 10 seconds.

Research and Development Engineer

PT Cipta Srigati Lestari | February 2011 to August 2011

Jakarta, Indonesia

- Developed Java-based optimization tools for B2B supply-chain and operational decision-support problems, translating business requirements into computational solutions.

PUBLISHED ARTICLES AND CONFERENCE PROCEEDINGS

Silalahi, F. T. R., **Saputra, D. W.**, Panjaitan, G., Butar-Butar, M., & Manurung, N. M. (2023). Influencing Student Choice: A Case Study of a Boarding Private University in Indonesia. *Proceedings of ICEBE 2022*, pp. 429-443. doi:10.2991/978-2-38476-064-0_46

Saputra, D. W. (2022). Optimalisasi Rute Distribusi Kurir Menggunakan Metode Traveling Salesman Problem (Studi Kasus: JNE Balige). *G-Tech: Jurnal Teknologi Terapan*, 6(2), 159-165. doi:10.33379/gtech.v6i2.1577

Meliala, G. N., & **Saputra, D. W.** (2020). Usulan Tata Letak Gudang Non Semen Menggunakan Metode Class-Based Storage di PT Intiaga Sukses Abadi Cabang Medan. *Jurnal Industri Kreatif*, 4(1), 17-26. doi:10.36352/jik.v4i01.23

Siboro, B. A. H., Sinaga, R. H. M., & **Devis Wawan Saputra Simanjuntak** (2020). Rancang Bangun Alat Pengering Andaliman dengan Menggunakan Metode Quality Function Deployment (QFD). *Jurnal Sains dan Teknologi*, 19(2), 133-140. Publication date: 2020 (ORCID). doi:10.36275/stsp.v19i2.210

Tampubolon, S. L., **Devis Wawan Saputra Simanjuntak**, & Simanjuntak, M. (2020). Analisis Kepuasan Wisatawan terhadap Atribut Wisata Menggunakan Metode CSI dan IPA pada Wisata Pemandian di Kabupaten Tapanuli Utara. *Jurnal Sains dan Teknologi*, 19(2), 141-151. Publication date: 2020 (ORCID). doi:10.36275/stsp.v19i2.216

Sinaga, R. H. M., & **Saputra, D. W.** (2019). Performance Investigation of Geothermal Power Plant Based on Exergy and Exergoeconomic Analyses (Case Study: Sarulla Geothermal Power Plant). *IOP Conference Series: Materials Science and Engineering*, 546, 072006. doi:10.1088/1757-899X/546/7/072006

Name variants: Devis Wawan Saputra, Devis W. Saputra and Devis Saputra. Publication names and years follow ORCID, including the recorded author variant “Devis Wawan Saputra Simanjuntak” for two articles.

BOOK, CURRICULUM AND INTELLECTUAL PROPERTY

Saputra, D., Siboro, B. A. H., & Sinaga, C. (2022). *Metodologi Pelaksanaan Kerja Praktik di Bidang Manajemen Rekayasa [Methodology for Conducting Practical Work in Engineering Management]*.

Saputra, D. (2019). *Curriculum of the Engineering Management Program*, Institut Teknologi Del, 2019 to 2024. Internal academic document.

Andaliman dryer developed using Quality Function Deployment: co-contributor to the applied research and associated registered copyright listed in ORCID (2020).

RESEARCH AND TECHNICAL COMPETENCE

Learning design and development: Training needs analysis; ADDIE and SAM; adult learning; learning objectives; curriculum and learning-path design; competency-based assessment; blended and experiential learning.

Learning systems and governance: Moodle implementation and operations; SCORM; course taxonomy; content audit and standardization; subject-matter expert collaboration; trainer coordination; learning-content ownership and maintenance.

Research and evaluation: Case studies; research synthesis; qualitative analysis; descriptive and comparative statistics; baseline comparisons; model calibration; error analysis; uncertainty and limitations; reproducible reporting.

AI engineering: developing through projects: Python, NumPy, pandas, scikit-learn and PyTorch; classification; sequence models; retrieval evaluation; learner modeling; test-set separation; leakage checks; Git and GitHub Actions.

Supporting tools: Articulate 360 (Rise and Storyline), Camtasia, Excel, SQL, Java, HTML and CSS. Familiarity with enterprise LMS environments includes Workday Learning, SAP SuccessFactors and Cornerstone OnDemand.

PROFESSIONAL DEVELOPMENT AND LANGUAGES

Advanced Machine Learning and Deep Learning, Indonesia AI (2025); Intercultural Team-Building Workshop, Gayfié, France (2013); German B2, Hartnackschule Berlin (2012).

Indonesian and Batak: native. English: working proficiency; TOEFL iBT 101/120 achieved in 2012, reported as historical evidence. German: B2 training (2012).

ACADEMIC SERVICE

Internal reviewer for journal and conference submissions (2016 to 2023); undergraduate thesis and capstone examiner; Engineering Management departmental senate member (2016 to 2018). Contributed to accreditation documentation, curriculum governance and internship processes.